Fortuitous Q-cohomology and quantum lifting: master report

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Contents

0. Conventions 1

1. $\mathcal{N}=4$, $SU(N)$ 2

2. $\mathcal{N}=4$, $SO(2N+1)$, $Sp(N)$ 5

3. $\mathcal{N}=2$ $\mathbb{Z}_2$ quiver 8

4. $\mathcal{N}=2$ SCQCD 10

5. Massive-adjoint flows 11

6. $SU(3)$ β / Leigh–Strassler 13

7. Running 16

0. Conventions

Spectral sequence

$$\mathbf{Q}=\sum_{n\geq 0}\hbar^n Q_n,\qquad \sum_{j=0}^{n} Q_j Q_{n-j}=0,\qquad E_1=H(Q_0)$$

$$\lambda=(\theta_1,\theta_2,\theta_3;n_+,n_-),\qquad \Theta=\theta_1+\theta_2+\theta_3,\qquad Y=\#\text{letters},\qquad \mathsf{d}=(1,1,1;-1,-1)$$

$$e_I=\text{unit vector of }\theta_I,\qquad e_\pm=\text{unit vector of }n_\pm$$

$$Q_n:\ C^Y(\lambda)\to C^{Y+1-n}(\lambda-n\mathsf{d}),\qquad d_r:\ E_r^Y(\lambda)\to E_r^{Y+1-r}(\lambda-r\mathsf{d})$$

$$\text{outgoing } d_r:\ r\leq \min_I \theta_I,\qquad \text{incoming } d_r:\ r\leq \min(n_+,n_-)$$

$$d_2[\mathcal{O}_0]=[Q_2\mathcal{O}_0+Q_1\mathcal{O}_1],\qquad Q_0\mathcal{O}_1=-Q_1\mathcal{O}_0$$

$$\mathcal{O}(\hbar)=\sum_{r\geq 0}\hbar^r\mathcal{O}_r,\qquad Q_0\mathcal{O}_r=-R_r,\qquad R_r=\sum_{k=1}^{r}Q_k\mathcal{O}_{r-k}$$

$$\operatorname{rk} d_R=\big(\operatorname{rk} M_R-\operatorname{rk} M_{R-1}\big)-\big(\operatorname{rk} M'_R-\operatorname{rk} M'_{R-1}\big)$$

$$M_R=\Big[\ \underbrace{Q_k\mathcal{K}}_{\text{cocycles}}\ \Big|\ Q_{k-m}:C_m\to C'_k\ \Big]_{1\leq m\leq k\leq R},\qquad M'_R=M_R\setminus\{Q_k\mathcal{K}\}$$

$$C_m=C^{Y-m}(\lambda-m\mathsf{d}),\qquad C'_k=C^{Y-k+1}(\lambda-k\mathsf{d})$$

Q_n on matrix letters

$$Q_nF=\mathrm{CENG}[n]\,\mu^{-n}\sum_{(i_1,\dots,i_{n+1})}\varepsilon\,A_nK_n\,F\big(\dots,\underbrace{[P,t_{R_1}]}_{i_1},\dots,\underbrace{[t^{R_1},t_{R_2}]}_{i_2},\dots,\underbrace{[t^{R_n},Q]}_{i_{n+1}},\dots\big)$$

$$\mu=-2,\qquad \mathrm{CENG}=(1,1,2,4,8),\qquad t^R=\kappa^{RS}t_S,\qquad \kappa_{RS}=\operatorname{tr}_F(t_Rt_S)$$

Charges

$$E=\tfrac12(\Theta+Y)+n_++n_-,\qquad R_I=\theta_I-\tfrac12(\Theta-Y),\qquad J_L=\tfrac12(\Theta-Y+n_++n_-),\qquad J_R=\tfrac12(n_+-n_-)$$

$$E=R_1+R_2+R_3+2J_L,\qquad 2(E+J_L)=2\Theta+3(n_++n_-)$$

$$(E,R_I,J_L,J_R)(\lambda+k\mathsf{d},Y+k)=(E,R_I,J_L,J_R)(\lambda,Y),\qquad (E;R_I;J_L,J_R)(\mathbf{Q})=(\tfrac12;\tfrac12,\tfrac12,\tfrac12;-\tfrac12,0)$$

twisted letter	physical letter	θ	jets	E	(R_1,R_2,R_3)	J_L	J_R
$(\gamma^I)_A$	$\bar\phi_I{}^A$	e_I	–	1	e_I	0	0
$(\beta_I)^A$	$-\frac{i}{\sqrt2}\psi_+^{IA}$	$(1,1,1)-e_I$	–	$\frac32$	$(\frac12,\frac12,\frac12)-e_I$	$\frac12$	0
$\delta^{AB}\mathsf{b}_B$	$\frac14 f^A_{++}$	$(1,1,1)$	–	2	$(0,0,0)$	1	0
$\partial_{+\dot a}\mathsf{c}^A$	$2\bar\lambda^A_{\dot a}$	0	$n_\pm=1$	$\frac32$	$(\frac12,\frac12,\frac12)$	0	$\pm\frac12$
$\partial_{+\dot a}$	$\nabla_{+\dot a}$	0	$n_\pm=1$	1	$(0,0,0)$	$\frac12$	$\pm\frac12$

Quantities

$$h = \dim H^Y(Q_0) \text{ on highest-weight cochains of } (\lambda, Y), \qquad g = \#\text{graviton classes}$$
$$\rho = \mathrm{rk}\big(\mathrm{res}_{\mathfrak{h}} : H^Y \to \text{Cartan restriction}\big)$$
$$0 \to R_N \to C_\infty \xrightarrow{\pi_N} C_N \to 0, \qquad f = h - \mathrm{rk}\,\pi_N^*$$
$$K(Y) = \dim\ker\big(H^Y(R_N) \to H^Y(C_\infty)\big) = f(Y-1)$$

Certification [$\mathbb{Q}$] exact over $\mathbb{Q}$; [p] ranks mod $p = 2^{31}-1$; [p,p′] also $p' = 2147483629$; [cond] $\mathrm{res}(R_r) = 0$ at targets with $h = \rho$.
All orders $\mathbf{Q}\,\mathcal{O}(\hbar) = 0$ (closed completion) $\wedge$ $\mathcal{O}_0 \notin \mathrm{im}\,d_r\ \forall r$.

1. $\mathcal{N} = 4$ Yang–Mills, $G = SU(N)$

1.1 Action and components

$$C^A(\theta) = ig\,\mathsf{c}^A + \theta_I \delta^{AB}(\gamma^I)_B - \tfrac{1}{2}\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A + \frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B, \qquad Q_+^I C^A = \frac{\partial_L C^A}{\partial\theta_I}$$
$$A = 1, \dots, N^2-1$$
$$S_{\mathrm{tw},E} = \int_{\mathbb{C}^2}\Omega \wedge \Big[\int d^3\theta\Big(\tfrac{1}{2}\delta_{AB}C^A\bar\partial C^B + \tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)}, \qquad Q_0 C^A = \tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$\begin{aligned} S_{\mathrm{tw},E} = \int_{\mathbb{C}^2}\Omega \wedge \Big[&\mathsf{b}_A\big(\bar\partial\mathsf{c}^A - \tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big) \\ &+ \sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B - ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B \\ &+ \tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F - \tfrac{1}{4}\mathsf{w}_{\dot a}^A\mathsf{w}^{A\dot a}\Big]_{(0,2)} \end{aligned}$$
$$(t_A^{\mathrm{adj}})^C{}_B = iC^C{}_{AB}, \qquad [t_B, t_C] = iC^A{}_{BC}t_A, \qquad \delta_{AB} = 2\,\mathrm{tr}(t_A t_B)$$
$$\mathsf{w}_{\dot a}^A \mapsto \tfrac{1}{2}\partial_{+\dot a}\mathsf{c}^A, \qquad -\tfrac{1}{4}\mathsf{w}_{\dot a}^A\mathsf{w}^{A\dot a} = -\tfrac{1}{16}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^A)$$
$$\begin{aligned}\mathcal{I} = &-\tfrac{g}{2}C^A{}_{BC}\mathsf{b}_A\mathsf{c}^B\mathsf{c}^C - ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C(\beta_I)^B \\ &+ \tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F - \tfrac{1}{4}\mathsf{w}_{\dot a}^A\mathsf{w}^{A\dot a}\end{aligned}$$

components: $\underbrace{\mathsf{c}^A,\ \mathsf{b}_A}_{\text{vector}}$, $\underbrace{(\gamma^I)_A,\ (\beta_I)^A}_{I=1,2,3\ \text{adjoint chirals}}$, C_{ABC} of $\mathfrak{su}(N)$; letters $X^{(r,s)} = \partial_{+\dot1}^r\partial_{+\dot2}^s X$, c only with $r+s \geq 1$

1.2 Lie algebra realization

$SU(2)$	basis $(e,h,f) = (E_{12},\ E_{11}-E_{22},\ E_{21})$	$[e,h] = -2e,\ \ [e,f] = h,\ \ [h,f] = -2f$
$SU(3)$	$(e_1,e_2,e_3,h_1,h_2,f_1,f_2,f_3)$	$= (E_{12},E_{23},E_{13},E_{11}-E_{22},E_{22}-E_{33},E_{21},E_{32},E_{31})$
	letter $A = \sum_X A^X t_X,\ \ [t_Y,t_Z] = C^X{}_{YZ}t_X$	$D_Z A^X = C^X{}_{ZY}A^Y,\ \ \kappa_{XY} = \mathrm{tr}(t_X t_Y)$
	singlets	$\mathrm{wt} = 0\ \cap\ \ker D_{f_1} \cap \ker D_{f_2}\quad (SU(2)\text{: } \mathrm{wt} = 0 \cap \ker D_f)$
	charge conjugation	$Ct_X = -t_X^{\mathsf{T}},\ \ C = \pm$ split $\frac{1}{2}(1 \pm C)$
$SU(N)$	fundamental $N \times N$, tr $= 0$	$\{E_{ij}\}_{i\neq j} \cup \{E_{ii} - E_{i+1,i+1}\},\ \ t^R = \kappa^{RS}t_S$
	Casimir tensor	$\sum_R (t_R)_{ab}(t^R)_{cd} = \delta_{ad}\delta_{bc} - \frac{1}{N}\delta_{ab}\delta_{cd},\ \ C_F = \frac{N^2-1}{N}$
	probes	$x \in \mathfrak{sl}_N(\mathbb{F}_p)$: N^2-1 random entries, $x_{NN} = -\sum_{i<N} x_{ii}$; $\mathrm{Tr}\,\mathbf{1} = N$
	trace words ($N \geq 4$)	$C_\infty = \mathbb{C}[\mathrm{tr}(w)]_{\lvert w\rvert\geq 2},\ \ R_N = \ker\big(\pi_N : C_\infty \to C_N\big)$ (trace identities)

1.3 Highest-weight algorithm

(a) $SU(2)$, $SU(3)$: bosonic highest weights

$$G_{\mathrm{HW}} = GL(3)_\theta \times SL(2)_{\dot a} \ \text{(letter-wise)}, \qquad [G_{\mathrm{HW}}, Q_n] = 0$$

$$\theta(\gamma^I) = e_I, \quad \theta(\beta_I) = (1,1,1) - e_I, \quad \theta(\mathsf{b}) = (1,1,1), \quad \theta(\mathsf{c}) = 0; \qquad (n_+, n_-)\big(X^{(r,s)}\big) = (r,s)$$

$$\lambda \text{ dominant}: \ \theta_1 \ge \theta_2 \ge \theta_3 \ge 0, \qquad n_+ \ge n_- \ge 0$$

$$C^Y_{\mathrm{HW}}(\lambda) = \big\{\text{colour singlets, weight } \lambda,\ Y \text{ letters}\big\} \cap \ker E_{12} \cap \ker E_{23} \cap \ker J_+, \qquad J_+ = u\,\partial_v \ \text{ on } \ X^{(r,s)} \leftrightarrow u^r v^s$$

$$C^Y_{\mathrm{HW}}(\lambda) = \bigoplus_{\text{skeletons}} C^Y_{\mathrm{HW}}(\lambda; S), \qquad S = \{(\text{type}, r+s)\} \text{ multiset}$$

$$\dim C^Y_{\mathrm{HW}}(\lambda; S) = \text{Weyl-alternated colour-singlet character (exact)}$$

$$SU(3): \quad g = \mathrm{Mult}\big(t_F \otimes t_B \otimes \tau\big), \qquad t_F: \ \gamma \mapsto V, \ \beta_I \mapsto e_{I+1} \wedge e_{I+2}, \ \mathsf{b} \mapsto e_1 \wedge e_2 \wedge e_3, \ \mathsf{c} \mapsto 1 \ \text{(column wedges of } \theta)$$

$$t_B = \prod_{n_-} [pq] \prod u, \quad [pq] = u_p v_q - v_p u_q, \qquad \tau = \text{colour trace word}, \qquad (g \pm Cg)(x_k) \text{ at probes } \Rightarrow V^{\pm}$$

$$h = \dim C^Y_{\mathrm{HW}} - \mathrm{rk}\, Q_0|_{Y-1} - \mathrm{rk}\, Q_0|_Y, \qquad \rho: \ \text{evaluation at diagonal probes}$$

(b) $SU(N \ge 4)$: $\mathfrak{gl}(3|2)$ super highest weights

$$V = \mathbb{C}^{3|2} = \langle \theta_1, \theta_2, \theta_3 \,|\, z_1, z_2 \rangle, \qquad \text{letters of degree } k: \ \Pi^{k-1}\Lambda^k V, \qquad L = \Theta + n_+ + n_-$$

$$\mu \vdash L, \quad \mu_4 \le 2 \ \ ((3|2) \text{ hook})$$

$$\lambda(\mu) = \big(\mu_1, \mu_2, \mu_3; \ \#\{i \ge 4 : \mu_i \ge 1\}, \ \#\{i \ge 4 : \mu_i \ge 2\}\big)$$

$$\text{raising: } E_{12}, \ E_{23}, \ J_+ \ (\text{even}), \quad \theta_f\,\partial_z \ (\text{odd}), \qquad \text{distinguished Borel}$$

$$K(\mu, Y) = \dim \ker\big(H^Y(R_N)_\mu \to H^Y(C_\infty)_\mu\big)$$

$$K = \dim R^Y - \mathrm{rk} \begin{pmatrix} Q: R^Y \to R^{Y+1} & 0 \\ \iota: R^Y \to C^Y_\infty & Q_0: C^{Y-1}_\infty \to C^Y_\infty \end{pmatrix} + \mathrm{rk}\, Q_0\big|_{C^{Y-1}_\infty} - \mathrm{rk}\, Q\big|_{R^{Y-1}}$$

stable range: $R^j_N = 0 \ \ (j < 2N+2)$; routes: $\ell(\mu) \le 2$, hooks: theorems; else exact Q and stable inclusion

1.4 Frontier

G	all sectors complete	partial / targeted
$SU(2)$	$E + J_L \le \frac{33}{2}$ [p]	detector all sectors $\le \frac{35}{2}$; $\frac{37}{2}$–22 partial; chains at $E + J_L = \frac{39}{2}, 20$
$SU(3)$	$E + J_L \le 15$ [p] ($\le \frac{23}{2}$: 2610.20340)	detector ≤ 17; windows at 16, $\frac{33}{2}$, 17, $\frac{35}{2}$
$SU(4)$	$E + J_L \le \frac{27}{2}$ [p]	14: 34/38 shapes (incl. $K = 1$)
$SU(5)$	≤ 14	$\frac{29}{2}$: 26/29; 15: 3/53
$SU(6)$	≤ 15	$\frac{31}{2}$: 1/38
$SU(7)$	≤ 15	$\frac{31}{2}$: 19/37; 16: 14/62
$SU(8)$	$\le \frac{31}{2}$	16: 45/56; $\frac{33}{2}$: 19/45
$SU(9)$	$\le \frac{33}{2}$	17: 41/64; $\frac{35}{2}$: 18/50
$SU(10)$	≤ 17	$\frac{35}{2}$: 33/38; 18: 39/72
$SU(11..15)$	≤ 18	–

1.5 Fortuitous generators and classes

G	class	λ	Y	E	(R_1, R_2, R_3)	J_L	J_R	data
$SU(2)$	$\mathcal{O}_0$	$(4,4,4;0,0)$	7	$\frac{19}{2}$	$(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{5}{2}$	0	primary; 2209.06728
	$\mathcal{O}_0$ multiplet	$E + J_L \in [\frac{25}{2}, \frac{33}{2}]$	7					69 HW blocks (descendants)
	$\mathcal{O}_1$	$(6,6,6;0,0)$	9	$\frac{27}{2}$	$(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{9}{2}$	0	primary; 2304.10155
	products	$E + J_L \in [16, \frac{33}{2}]$	9					graviton $\times \mathcal{O}_0$, descendants: fresh 0
	P39	$(5,5,5;3,0)$	9	15	$(2,2,2)$	$\frac{9}{2}$	$\frac{3}{2}$	$h = 13$, known 12, fresh 1 [p,p′]
	X40	$(7,5,5;1,1)$	10	$\frac{31}{2}$	$(\frac{7}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{9}{2}$	0	$(h, \rho, g) = (8, 8, 7)$ [p,p′]
$SU(3)$	X_7	$(2,2,2;2,2)$	7	$\frac{21}{2}$	$(\frac{5}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{3}{2}$	0	$C = -1$, $(h, \rho, g) = (26, 26, 25)$; 35 HW blocks
	X_7 member	$(0,0,0;5,5)$	7	$\frac{27}{2}$	$(\frac{7}{2}, \frac{7}{2}, \frac{7}{2})$	$\frac{3}{2}$	0	fortuitous [$\mathbb{Q}$]; 26 trace generators
	F_1 tower	$(n{+}4,4,4;0,0)$	$n{+}7$	$n{+}\frac{19}{2}$	$(n{+}\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{5}{2}$	0	$n \ge 3$; 2412.08695, 2610.20340
	B_1 tower	$(n{+}5,5,4;0,0)$	$n{+}8$	$n{+}11$	$(n{+}2, 2, 1)$	3	0	2412.08695, 2610.20340
	new primary	$(5,4,4;1,1)$	10	$\frac{27}{2}$	$(\frac{7}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{5}{2}$	0	$C = -1$, $h = 2$, $\rho = 0$ [cond: free]

G	class	λ	Y	E	(R_1,R_2,R_3)	J_L	J_R	data
	new primary	$(7,4,4;1,0)$	11	14	$(5,2,2)$	$\frac{5}{2}$	$\frac{1}{2}$	$C=+1$, $h=2$ (F_1 desc.+1), $\rho=0$
	new primary	$(6,4,4;1,1)$	11	$\frac{29}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{5}{2}$	0	$C=+1$, $h=3$, $\rho=0$
	new primary	$(7,5,4;1,0)$	11	$\frac{29}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{3}{2})$	3	$\frac{1}{2}$	$C=+1$, $h=4$, $\rho=0$
$SU(4)$	first class	$(4,2,2;2,2)$	9	$\frac{25}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{3}{2}$	0	$C=-1$, $K=1$ [$\mathbb{Q}$]; first $SU(4)$ class
	(Codex)	$(2,2,2;3,3)$	9	$\frac{27}{2}$	$(\frac{7}{2},\frac{7}{2},\frac{7}{2})$	$\frac{3}{2}$	0	17500-term stable primitive; $=(4,2,2;2,2)-2e_1+e_+ +e_-$
$SU(5..15)$	–							none through the frontier

descendant shifts at fixed Y : $(-e_I;+e_\pm)\ (\theta\,\partial_z)$, $(+e_I;0)\ (\partial_\theta)$, $(7,4,4;1,0)Y11 \to (6,4,4;1,1)Y11$: $(-e_1;+e_-)$

1.6 Quantum lift

class	result	level
$\mathcal{O}_0$ ($SU(2)$)	closed completion $\sum_{r\le 4}\hbar^r\mathcal{O}_r$; no incoming $\Rightarrow$ all orders	[$\mathbb{Q}$]
$\mathcal{O}_1$ ($SU(2)$)	$d_1\neq 0$ (2606.27955)	[p,p$'$]
P39	$d_1=0$; $R_2,R_3,R_4\in\operatorname{im}Q_0$; $r=5$ target $h=0$; no incoming $\Rightarrow$ all orders	[p,p$'$]
X40	$d_1=0$; $R_2\in\operatorname{im}Q_0+Q_1H$; $R_3,R_4\in\operatorname{im}Q_0$; $[R_5]=0$; incoming $h=0\Rightarrow$ all orders	[p,p$'$]
		($r\le 2$), [p]
X_7	$d_1=0$ in/out; d_2 out target $h=0$, in source $h=0$; $r\ge 3$ empty $\Rightarrow$ all orders	[p]
F_1, $n=3$	closed completion $\mathcal{O}_0..\mathcal{O}_4$; $r\ge 5$ empty; no incoming $\Rightarrow$ all orders ($d_1=0$: 2610.20340)	[p]
$(5,4,4;1,1)Y10$	$d_1=d_3=0$, $d_4=0$ [cond], no incoming; d_2: target $(3,2,2;3,3)Y9$, $h=331$, $\rho=329$	running
$(7,4,4;1,0)Y11$	$d_1=0$ (target $h=0$), $d_2=0$ [cond] ($h=\rho=1464$), $d_3=0$, $d_4=0$ [cond], no incoming $\Rightarrow$ all orders	[p], [cond]
$(6,4,4;1,1)Y11$	no incoming; $d_3=0$; $d_4=0$ [cond]; d_1, d_2 open	[p]
$(7,5,4;1,0)Y11$	no incoming; $d_4=0$ [cond] ($h=\rho=501$); d_1,d_2,d_3 open	[p]
B_1, $(8,5,4;0,0)Y11$	d_1 target $(7,4,3;1,1)Y11$: $h=2$, $\rho=0$	open
$SU(4)$, $(4,2,2;2,2)Y9$	d_1: R_1 exact at $(3,1,1;3,3)Y9^-$ ($h=111$); R_2 exact ($h=0$); $r\ge 3$ empty; incoming d_1 rank 0 (453 sources), d_2 source $h=0\Rightarrow$ all orders	[p]
(Codex) $(2,2,2;3,3)Y9$	five Q_1 scalar queries $=0$	partial

Chain data: P39, $(5,5,5;3,0)Y9$

r	target	target data	verdict
1	$(4,4,4;4,1)$, 9	$h=13$	$d_1=0$
2	$(3,3,3;5,2)$, 8	$g_8=583$	$R_2\in\operatorname{im}Q_0$ (45,450,714 terms)
3	$(2,2,2;6,3)$, 7	$h=1$	$R_3\in\operatorname{im}Q_0$ (17,787,235)
4	$(1,1,1;7,4)$, 6	$h=\rho=g=152$	$R_4\in\operatorname{im}Q_0$ (1,276,152)
5	$(0,0,0;8,5)$, 5	$h=0$	–

r	X40 target	data	verdict
1	$(6,4,4;2,2)$, 10	–	$d_1=0$ (4,429,603)
2	$(5,3,3;3,3)$, 9	–	$R_2\in\operatorname{im}Q_0+Q_1H$
3	$(4,2,2;4,4)$, 8	$g_8=1112$	$R_3\in\operatorname{im}Q_0$
4	$(3,1,1;5,5)$, 7	–	$R_4\in\operatorname{im}Q_0$
5	$(2,0,0;6,6)$, 6	$h=\rho=g=30$	$\operatorname{res}R_5=0$

Chain data: F_1, $n=3$, $(7,4,4;0,0)Y10$, $C=-1$

r	target	dim /h/ρ	residual	$\mathcal{O}_r$ (coordinates; $\operatorname{rk}Q_0$)
1	$(6,3,3;1,1)$, 10	34430/0/0	$Q_1\mathcal{O}_0$	$Y9$ (28311; 18003)
2	$(5,2,2;2,2)$, 9	63983/609/608	$Q_2\mathcal{O}_0+Q_1\mathcal{O}_1$	$Y8$ (49328; 32144)
3	$(4,1,1;3,3)$, 8	26400/0/0	$Q_2\mathcal{O}_1+Q_1\mathcal{O}_2$	$Y7$ (18919; 13063)
4	$(3,0,0;4,4)$, 7	1896/34/34	$Q_3\mathcal{O}_1+Q_2\mathcal{O}_2+Q_1\mathcal{O}_3$	$Y6$ (1169; 825)

r	$SU(4)$ $(4,2,2;2,2)Y9$: target, $C=-1$	data
1	$(3,1,1;3,3)$, 9	18133/12574/5448, $h=111$
2	$(2,0,0;4,4)$, 8	983/666/317, $h=0$
in 1	$(5,3,3;1,1)$, 9	$h=453$, $\operatorname{rk}d_1=0$
in 2	$(6,4,4;0,0)$, 10	$h=0$

Structural

$$\mathrm{res}(\hbar)\,\mathbf{Q}=\mathbf{Q}_{\rm IR}\,\mathrm{res}(\hbar),\quad \mathbf{Q}_{\rm IR}=0\ \Rightarrow\ \mathrm{res}(R_r)=0\ \Rightarrow\ d_r=0 \text{ at targets with } h=\rho\quad([\text{cond}];\ r=1:\ 2610.22332)$$

$$SU(2),\ Y=11,\ E+J_L\in[\tfrac{41}{2},\tfrac{45}{2}]:\ \ 7 \text{ blocks},\ \dim E_2=17,\ E_2\subset\mathrm{span}(E_2\cdot E_2)$$

$$\rho-g=1:\ \text{X40},\ X_7\quad(\text{Coulomb-visible fortuitous})$$

$$0\neq[X_7]\in H^7\big(\mathfrak{sl}_3[z_+,z_-],\mathfrak{sl}_3;\mathbb{C}\big)_{(5,5)},\qquad X_7(z)=4838400\quad[\mathbb{Q}]$$

1.7 Data

$SU(2)$ per level states $=\sum h\cdot\dim(\theta)_{SU(3)}\cdot(n_+-n_-+1)$

$E{+}J_L$	sect.	HW	B	F	NG blk	NG	$E{+}J_L$	sect.	HW	B	F	NG blk	NG
1	1	0	0	0	0	0	9	39	112	1786	1173	0	0
$\frac{3}{2}$	1	0	0	0	0	0	$\frac{19}{2}$	26	140	1542	2934	0	0
2	2	1	6	0	0	0	10	46	211	3327	3507	0	0
$\frac{5}{2}$	1	1	0	6	0	0	$\frac{21}{2}$	39	271	6618	3734	0	0
3	5	2	1	8	0	0	11	59	413	6807	8733	0	0
$\frac{7}{2}$	2	2	18	0	0	0	$\frac{23}{2}$	46	531	9582	13824	0	0
4	6	4	18	12	0	0	12	81	811	21483	13593	1	1
$\frac{9}{2}$	5	4	2	38	0	0	$\frac{25}{2}$	59	1053	26406	25614	1	1
5	9	6	36	30	0	0	13	94	1563	30504	46380	3	3
$\frac{11}{2}$	6	7	102	18	0	0	$\frac{27}{2}$	81	2028	63444	49640	4	5
6	16	15	59	139	0	0	14	116	2986	90111	74934	7	9
$\frac{13}{2}$	9	17	96	228	0	0	$\frac{29}{2}$	94	3835	101250	138786	7	11
7	19	29	423	114	0	0	15	150	5567	177657	170608	12	19
$\frac{15}{2}$	16	37	402	420	0	0	$\frac{31}{2}$	116	7203	280314	222792	9	19
8	26	60	345	912	0	0	16	172	10330	333117	391821	14	27
$\frac{17}{2}$	19	72	1230	714	0	0	$\frac{33}{2}$	150	13377	494548	548270	15	30

NG = odd-Y (non-graviton) highest-weight classes: all in the $\mathcal{O}_0$ multiplet at $Y=7$; $Y=9$ blocks at $E+J_L=16,\frac{33}{2}$: products/descendants (fresh 0).

$SU(N\geq 4)$ relation scan

$$\text{shape } \mu \text{ at level } 2(E+J_L)=2(\mu_1+\mu_2+\mu_3)+3\textstyle\sum_{i\geq 4}\mu_i$$

K one prime; $K(4,2,2,2,2)=1$ $[\mathbb{Q}]$, all other completed shapes: $K=0$

$SU(4)$, $\mu=(4,2,2,2,2)$, Y	5	6	7	8	9	10	11	12
$\dim R^Y$	17	213	1201	3430	5321	4579	2085	404
$\mathrm{rk}\,Q\vert_{R^Y}$	17	196	1005	2424	2894	1681	401	0
$\dim H^Y(R_4)$	0	0	0	1	3	4	3	3
K	0	0	0	0	0	**1**	0	0

$K=4579-7900+6216-2894=1$ $[\mathbb{Q}:\ 359/467/783 \text{ primes} > \text{Hadamard}]$, $SU(5),\ \mu=(4,2,2,2,2):\ K=0$

2. $\mathcal{N}=4$ Yang–Mills, $G=SO(2N+1),\ Sp(N)$

2.1 Action and components

$$C^A(\theta)=ig\,\mathsf{c}^A+\theta_I\delta^{AB}(\gamma^I)_B-\tfrac{1}{2}\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A+\frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B,\qquad Q^I_+C^A=\frac{\partial_L C^A}{\partial\theta_I}$$

$$A=1,\dots,\dim G=N(2N+1)$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\int d^3\theta\Big(\tfrac{1}{2}\delta_{AB}C^A\bar\partial C^B+\tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)},\qquad Q_0C^A=\tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+\tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$
$$(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\qquad[t_B,t_C]=iC^A{}_{BC}t_A,\qquad\delta_{AB}=2\,\mathrm{tr}(t_At_B)$$
$$\mathsf{w}^A_{\dot a}\mapsto\tfrac{1}{2}\partial_{+\dot a}\mathsf{c}^A,\qquad-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}=-\tfrac{1}{16}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^A)$$
$$C_{ABC}\text{ of }\mathfrak{so}(2N+1)\text{ or }\mathfrak{sp}(N)$$
$$\text{components: }\mathsf{c}^A,\ \mathsf{b}_A\text{ (vector)},\quad(\gamma^I)_A,\ (\beta_I)^A\ (I=1,2,3\text{ adjoint chirals})$$

2.2 Lie algebra realization

$$SO(2n+1):\ N_F=2n+1,\quad A^{\mathsf T}\eta+\eta A=0,\quad\eta_{i\bar\imath}=1;\qquad Sp(n):\ N_F=2n,\quad A^{\mathsf T}\Omega+\Omega A=0,\quad\Omega_{i\bar\imath}=s_i$$
$$\bar\imath=N_F+1-i,\qquad s_i=+1\ (i\le n),\quad s_i=-1\ (i>n),\qquad\mathfrak{h}=\text{diagonal}$$
$$\text{basis: independent subset of }\ E_{ij}-E_{\bar\jmath\bar\imath}\ (SO),\qquad E_{ij}-s_is_jE_{\bar\jmath\bar\imath}\ (Sp)$$
$$\kappa_{RS}=\mathrm{tr}_F(t_Rt_S),\qquad\sum_R(t_R)_{ab}(t^R)_{cd}=\tfrac{1}{2}\delta_{ad}\delta_{bc}-\tfrac{1}{2}F_{ac}F_{bd},\quad F=\eta\text{ or }\Omega,\qquad C_F=\tfrac{N_F-1}{2}\ (SO),\ \tfrac{N_F+1}{2}\ (Sp)$$

trace words: $O(\infty)$ and $Sp(\infty)$ stable algebras coincide (reversal identity); C_∞ realized on $SO(n)$, $n\ge\#\text{letters}$

2.3 Highest-weight algorithm

$$G_{\mathrm{HW}},\ \lambda\text{ dominant},\ C^Y_{\mathrm{HW}}(\lambda)\text{ as 1.3(a)}$$
$$\dim C^Y_{\mathrm{HW}}:\ \text{Weyl integration, torus points summed over signed-permutation orbits (exact)}$$
$$\dim C^Y_\infty:\ \text{Burnside over }\prod S_2\wr S_k\text{ on perfect matchings}$$
$$f_Y=h_Y-\mathrm{rk}\,\pi^*,\qquad\mathrm{rk}\,\pi^*=\mathrm{rk}\begin{pmatrix}\Pi & B_G\\ Q_0^\infty & 0\end{pmatrix}-\mathrm{rk}\,Q_0^\infty-\mathrm{rk}\,B_G$$
$$\text{stable range: }Y<N_F+1\ (SO),\quad Y<n+1\ (Sp(n))\ \Rightarrow\ f=0$$

2.4 Frontier

(one prime, all sectors)

G	complete $E+J_L\le$	partial (done/total sectors)
SO(3)	12	13: 16/94, 14: 11/116
Sp(1)	12	13: 16/94, 14: 11/116
SO(5)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
Sp(2)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
SO(7)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
Sp(3)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116
SO(9)	$\frac{23}{2}$	12: 17/81, 13: 16/94, 14: 11/116
Sp(4)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116
SO(11)	12	$\frac{25}{2}$: 17/59, 13: 48/94, $\frac{27}{2}$: 1/81, 14: 25/116
Sp(5)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116

$SO(3)=Sp(1)$: $SU(2)$ census ($\le\frac{33}{2}$).

2.5 Fortuitous generators and classes

G	class	λ	Y	E	(R_1,R_2,R_3)	J_L	J_R	data
$SO(3)=Sp(1)$	$\mathcal{O}_0$	$(4,4,4;0,0)$	7	$\frac{19}{2}$	$(\frac{3}{2},\frac{3}{2},\frac{3}{2})$	$\frac{5}{2}$	0	first level
$SO(5)=Sp(2)$	first class	$(3,2,2;1,1)$	7	9	$(3,2,2)$	1	0	first level; $f=1$, $\rho=0$
$SO(7)$	O_f	$(3,3,3;0,0)$	8	$\frac{17}{2}$	$(\frac{5}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	0	first level; 2506.13887; $(h,\rho)=(1,1)$
$Sp(3)$	first class	$(5,2,2;1,1)$	9	11	$(5,2,2)$	1	0	first level; $h=f=1$
$Sp(3)$	ω	$(5,0,0;3,3)$	9	13	$(7,2,2)$	1	0	$h=1$, $\rho=0$; $\lambda=(5,2,2;1,1)-2e_2-2e_3+2e_++2e_-$
$SO(9)$	first class	$(5,3,3;0,0)$	10	$\frac{21}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	0	first level; $h=6=5+1$
$SO(11)$	class	$(7,3,3;0,0)$	12	$\frac{25}{2}$	$(\frac{13}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	0	$h=23$, $f=1$ ($E+J_L=\frac{25}{2}$ partial)
$Sp(4),Sp(5)$	–							none through $E+J_L\le12$

$SO(2N+1),\ N=3,4,5:\quad \lambda=(2N-3,3,3;0,0),\quad Y=2N+2,\quad E=\frac{4N+5}{2},\quad J_L=\frac{1}{2}$

hits above the first level (2.7): shifts $(-e_I;+e_\pm),\ (+e_I;0)$ at fixed Y of the first class

2.6 Quantum lift

class	result	level
O_f $(SO(7))$	d_1 rank 1 into $O_m=(2,2,2;1,1)Y8$ $(h=2,\rho=1)$; 2511.09519	[p] (three primes)
$SO(7)$ $(4,3,3;0,0)Y8$	d_1 rank 1 into $(3,2,2;1,1)Y8$ $(h=11,\ \rho=10)$	[p]
$SO(7)$ $(3,1,1;2,2)Y8$	$d_1\neq 0$ into $(2,0,0;3,3)Y8$ $(h=2,\rho=1)$; $Sp(3)$: $d_1=0$	[$\mathbb{Q}$]
$SO(5)$ $(3,2,2;1,1)Y7$	out d_1, d_2 targets $h=0$; in d_1 source $h=0$; $r\geq 3$ empty $\Rightarrow$ all orders	[p]
$SO(9)$ $(5,3,3;0,0)Y10$	d_1 rank 1 into $(4,2,2;1,1)Y10$ $(h=10,\ \rho=9)$	[p]
$SO(11)$ $(7,3,3;0,0)Y12$	d_1,d_2,d_3 (engine with $N_F\leq 11$)	running
$Sp(3)$ $(5,2,2;1,1)Y9$	out $d_1=0$ $(Q_1\mathcal{K}\neq 0)$; in d_1: source $(6,3,3;0,0)Y9$ $h=0$; out $d_2=0$ (target $(3,0,0;3,3)Y8$, $h=6$); $r\geq 3$ empty $\Rightarrow$ all orders	[p]
ω	no outgoing; in d_1, d_3: sources $h=0$; in d_2: from $(7,2,2;1,1)Y10$ $(h=454)$	open
$SO(7)$	$h-\rho=1$, $f=0$ (monotone); in d_1 from $(7,2,2;1,1)Y10$: rank 0 (full recheck,	[p]
$\beta=(6,1,1;2,2)Y10$	$Q_1\mathcal{K}\neq 0$; $Sp(3)$: 0); in d_2 source $h=0$	

after one loop: $(h,\rho)^{SO(2N+1)}=(h,\rho)^{Sp(N)}$ in all four lifted sectors

S-duality test of $E_2=E_\infty$

$$E_2=E_\infty\ \wedge\ \text{S-duality}\ \Longrightarrow\ \sum_{(\lambda,Y)\mapsto(E,R_I,J_L,J_R)}\dim E_2^{SO(2N+1)}=\sum_{(\lambda,Y)\mapsto(E,R_I,J_L,J_R)}\dim E_2^{Sp(N)}\quad\forall\,(E,R_I,J_L,J_R)$$

mismatch at any charge $\Rightarrow$ $d_{r\geq 2}\neq 0$ in $SO(2N+1)$ or $Sp(N)$; $\quad SO(7),\ Sp(3):\ E+J_L\leq 12$ running

$SO(7)$ vs $Sp(3)$ per (E,R_I,J_L,J_R), all sectors [p]

$E+J_L$	5	$\frac{11}{2}$	6	$\frac{13}{2}$	7	$\frac{15}{2}$	8	$\frac{17}{2}$	9	$\frac{19}{2}$	10	$\frac{21}{2}$
$\sum h^{SO(7)}$	12	14	34	39	79	100	203	255	497	649	1236	1646
$\sum h^{Sp(3)}$	12	14	34	39	79	100	203	255	495	647	1230	1636
charges with $E_1^{SO(7)}\neq E_1^{Sp(3)}$	0	0	0	0	0	0	0	0	2	2	6	7
$\#\{d_1\neq 0\}$: $SO(7)$ / $Sp(3)$	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	1/0	1/0	3/0	4/0
$\sum\dim E_2$: $SO(7)=Sp(3)$	12	14	34	39	79	100	203	255	495	647	1230	1636
charges with $E_2^{SO(7)}\neq E_2^{Sp(3)}$	0	0	0	0	0	0	0	0	**0**	**0**	**0**	**0**

d_2 candidates: none (both); $\quad d_3$ candidates: $(4,3,3;0,0)Y6, Y8$ $(E+J_L=10)$, 2 $(E+J_L=\frac{21}{2})$, rank 0 (both)

$E_\infty^{SO(7)}=E_\infty^{Sp(3)}$ at every charge, $E+J_L\leq\frac{21}{2}$

2.7 Data

h (computed sectors) / **f** per level $\quad$ * stable-range sectors not counted in h; ^{p} partial level

$E+J_L$	$SO(3)$	$Sp(1)$	$SO(5)$	$Sp(2)$	$SO(7)$	$Sp(3)$	$SO(9)$	$Sp(4)$	$SO(11)$	$Sp(5)$
8	60	60	164	164	43*	203	0*	207	0*	207
$\frac{17}{2}$	72	72	207	207	117*	255	0*	261	0*	261
9	112	112	381	381	385 / **1***	495	0*	516	0*	516
$\frac{19}{2}$	140	140	495	495	596 / **1***	647	0*	675	0*	675
10	211	211	877 / **1**	877 / **1**	1218 / **3***	1230	168*	1304	0*	1309
$\frac{21}{2}$	271	271	1151 / **2**	1151 / **2**	1641 / **5***	1636	554*	1736	0*	1745
11	413	413	1996 / **6**	1996 / **6**	3047 / **9**	3029	2058 / **1***	3277	0*	3308
$\frac{23}{2}$	531	531	2665 / **9**	2665 / **9**	4099 / **11**	4077	3627 / **2***	4426	0*	4470
12	811 / **1**	811 / **1**	90 / **1**^{p}	90 / **1**^{p}	100^{p}	7408 / **1**	0^{p}	8252	651*	8374
$\frac{25}{2}$	–	–	–	–	–	1385^{p}	–	1748^{p}	1642^{p}	1821^{p}
13	261^{p}	261^{p}	726 / **7**^{p}	726 / **7**^{p}	868^{p}	1380^{p}	842^{p}	1659^{p}	848 / **1**^{p}	1741^{p}
$\frac{27}{2}$	–	–	–	–	–	24^{p}	–	44^{p}	50^{p}	50^{p}
14	160^{p}	160^{p}	378 / **6**^{p}	378 / **6**^{p}	429^{p}	424^{p}	434^{p}	442^{p}	13^{p}	447^{p}

(h,ρ) of lifted pairs

sector	$SO(2N+1)$	$Sp(N)$
$(3,3,3;0,0)Y8$	$(1,1)$	$(0,0)$
$(2,2,2;1,1)Y8$	$(2,1)$	$(1,1)$
$(4,3,3;0,0)Y8$	$(7,7)$	$(6,6)$
$(3,2,2;1,1)Y8$	$(11,10)$	$(10,10)$
$(5,3,3;0,0)Y10$	$(6,6)$	$(5,5)$
$(4,2,2;1,1)Y10$	$(10,9)$	$(9,9)$

ω orbit	$SO(7)$ h/ρ	$Sp(3)$ h/ρ
$(5,0,0;3,3)Y9$	0/0	1/0
$(6,1,1;2,2)Y10$	355/354	354/354
$(7,2,2;1,1)Y11$	1/0	1/0
$(8,3,3;0,0)Y12$	41/40	41/40

3. $\mathcal{N}=2$ $\mathbb{Z}_2$ quiver, $G=SU(N)\times SU(N)$

3.1 Action and components

$$V=V_0\oplus V_1\subset\mathfrak{gl}(2N),\qquad V_0=\left\{\begin{pmatrix}a&0\\0&b\end{pmatrix}:\operatorname{tr}a=\operatorname{tr}b=0\right\}=\mathfrak{su}(N)\oplus\mathfrak{su}(N)$$

$$V_1=\left\{\begin{pmatrix}0&x\\y&0\end{pmatrix}\right\}=(\mathbf{N},\bar{\mathbf{N}})\oplus(\bar{\mathbf{N}},\mathbf{N})$$

$$t_A \text{ Hermitian basis of } V,\quad \operatorname{tr}(t_At_B)=\tfrac12\delta_{AB},\qquad C_{ABC}=-2i\operatorname{tr}\big(t_A[t_B,t_C]\big),\qquad \sigma=\operatorname{diag}(\mathbf{1}_N,-\mathbf{1}_N)$$

$$C^A(\theta)=ig\,\mathsf{c}^A+\theta_I\delta^{AB}(\gamma^I)_B-\tfrac12\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A+\frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B,\qquad Q^I_+C^A=\frac{\partial_LC^A}{\partial\theta_I}$$

$$A=1,\dots,\dim V$$

$$\sigma\,C(\theta_1,\theta_2,\theta_3)\,\sigma^{-1}=C(-\theta_1,-\theta_2,\theta_3)\ \Rightarrow\ \mathsf{c},\mathsf{b},\gamma^3,\beta_3\in V_0,\qquad \gamma^1,\gamma^2,\beta_1,\beta_2\in V_1$$

$$S_{\text{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\int d^3\theta\Big(\tfrac12\delta_{AB}C^A\bar\partial C^B+\tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)},\qquad Q_0C^A=\tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$g_1=g_2=g$$

$$\begin{aligned}S_{\text{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[&\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0}\\&+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\text{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B\\&+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}\end{aligned}$$

$$A\in V_0;\qquad B,C\in V_0\ (I=3),\qquad B,C\in V_1\ (I=1,2);\qquad D,E\in V_1,\ F\in V_0$$

$$(t_A^{\text{adj}})^C{}_B=iC^C{}_{AB},\qquad [t_B,t_C]=iC^A{}_{BC}t_A,\qquad \delta_{AB}=2\operatorname{tr}(t_At_B)$$

$$\mathsf{w}^A_{\dot a}\mapsto\tfrac12\partial_{+\dot a}\mathsf{c}^A,\qquad -\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}=-\tfrac{1}{16}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^A)$$

$$\text{components:}\qquad \underbrace{\mathsf{c},\mathsf{b}}_{\text{two vector multiplets}}\ ,\qquad \underbrace{\gamma^3,\beta_3}_{\text{two adjoint chirals}}$$

$$\underbrace{\gamma^1,\gamma^2,\beta_1,\beta_2}_{\text{two bifundamental hypermultiplets}}$$

$$\kappa=g_2/g_1:\quad Q_0\ni\kappa^2 \text{ on the node-2 block of the hypermultiplet moment maps}$$

$$Q_n \text{ weights: } \prod_{\text{inputs}} w\ \prod_{\text{vertices}} \text{coef}\ \prod_{\text{lines}} \text{prop}$$

$$\#\kappa^2 \text{ in a term of } Q_n=\#(\mathsf{b}_2,\beta_{3,2}\text{ consumed})-\#(\mathsf{b}_2,\beta_{3,2}\text{ produced})$$

$$\Rightarrow\ Q_n(\kappa)=\phi_\kappa Q_n(1)\phi_\kappa^{-1}\ \ \forall n\ \Rightarrow\ E_r(\kappa)\cong E_r(1)\ \ (\kappa\neq0)$$

3.2 Lie algebra realization

$$N_F=2N,\qquad \mathfrak{g}=\left\{\begin{pmatrix}a&0\\0&b\end{pmatrix}:\operatorname{tr}a=\operatorname{tr}b=0\right\},\qquad \text{bifundamental lines: } E_{ij}\ (i\le N<j \text{ or } j\le N<i),\ \text{dual } E_{ji}$$

$$\text{node-2 generators: } t^R\to\kappa^2t^R;\qquad \text{weights: } U(N)\times U(N)/U(1)_{\text{diag}} \text{ torus } \Rightarrow \text{ baryon-number-zero invariants}$$

3.3 Highest-weight algorithm

$$G_{\mathrm{HW}} = GL(2)_{12} \times GL(1)_3 \times SL(2)_{\dot{a}}, \qquad \theta_1 \geq \theta_2 \geq 0, \;\; \theta_3 \geq 0, \;\; n_+ \geq n_- \geq 0$$

$$C^Y_{\mathrm{HW}}(\lambda) = \{\text{colour singlets}\} \cap \ker E_{12} \cap \ker J_+, \qquad \text{columns: } \theta_1 \text{ wedge columns of } (\theta_1, \theta_2) \;\oplus\; \theta_3 \text{ single slots}$$

3.4 Frontier

(all sectors)

$$SU(2)^2: \; E + J_L \leq \tfrac{21}{2}; \qquad SU(3)^2: \; E + J_L \leq 9$$

3.5 Classes and quantum lift

Classes f: open (no stable comparison); E_2 classes per level in 3.6

$$E_2 = h - \mathrm{rk}\, d_1^{\mathrm{out}} - \mathrm{rk}\, d_1^{\mathrm{in}}; \qquad d_2 \text{ candidates: } E_2^Y(\lambda) > 0, \; E_2^{Y-1}(\lambda - 2\mathsf{d}) > 0$$

$$\mathrm{rk}\, d_2 = 0 \;\text{ on every candidate pair, }\; \kappa^2 = 1, 9$$

$$SU(2)^2: \; 37 \text{ pairs } (Q_2\mathcal{K} \neq 0: \; 29), \qquad SU(3)^2: \; 26 \text{ pairs } (Q_2\mathcal{K} \neq 0: \; 9), \qquad \max \mathrm{rk}\, d_2 = 0$$

$$d_3: \; (3,3,3;0,0)Y6 \to (0,0,0;3,3)Y4: \; \mathrm{rk} = 0 \;\; (Q_3\mathcal{K} = 0)$$

$$\text{sources } (E + J_L \leq 9): \; \rho = 0$$

$$\text{targets } (h, \rho): \; (1,1,0;2,2)Y4 \; (6,4), \; (2,0,0;2,2)Y4 \; (2,0), \; (1,1,1;2,2)Y4 \; (20,14), \; (2,0,1;2,2)Y4 \; (6,0)$$

$$\text{control: } Q_2 \to 4Q_2 \;\Rightarrow\; \mathrm{rk}\, d_2 = 3$$

3.6 Data

Columns

$$\text{HW blocks} = \#\{(\lambda, Y) : h > 0\}, \qquad \textstyle\sum h = \sum_{(\lambda,Y)} h$$

$$\#\{d_1 \neq 0\} = \#\{(\lambda, Y) : \mathrm{rk}\, d_1 > 0\}, \qquad \textstyle\sum E_2 = \sum h - 2 \sum \mathrm{rk}\, d_1$$

$$d_2 \text{ pairs} = \#\big\{(\lambda, Y) : E_2^Y(\lambda) > 0, \; E_2^{Y-1}(\lambda - 2\mathsf{d}) > 0\big\} \;\; (\text{where } d_2 \text{ can act})$$

$$\max \mathrm{rk}\, d_2 = 0 \;\Rightarrow\; \text{no two-loop lift}$$

$$\#\{Q_2\mathcal{K} \neq 0\} = \#\{\text{pairs with } Q_2 \neq 0 \text{ on the } E_2 \text{ cocycles}\} \;\; (\text{non-trivial tests})$$

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1 \neq 0\}$	$\sum E_2$	d_2 pairs	$\max \mathrm{rk}\, d_2$	$\#\{Q_2\mathcal{K} \neq 0\}$ $(\kappa^2 = 1, 9)$
2	2	4	3	4	0	4	0	–	–
2	$\frac{5}{2}$	2	1	2	0	2	0	–	–
2	3	8	4	6	0	6	0	–	–
2	$\frac{7}{2}$	4	4	6	0	6	0	–	–
2	4	13	11	17	0	17	0	–	–
2	$\frac{9}{2}$	8	7	14	0	14	0	–	–
2	5	20	19	38	1	36	0	–	–
2	$\frac{11}{2}$	13	15	42	1	40	0	–	–
2	6	31	34	93	2	89	0	–	–
2	$\frac{13}{2}$	20	26	116	1	112	0	–	–
2	7	44	57	240	3	230	1	0	0
2	$\frac{15}{2}$	31	43	276	3	268	1	0	0
2	8	61	90	560	5	544	3	0	4
2	$\frac{17}{2}$	44	73	716	4	696	1	0	2
2	9	84	140	1298	7	1258	7	0	10
2	$\frac{19}{2}$	61	110	1690	8	1640	5	0	8
2	10	111	205	3153	13	3059	11	0	18
2	$\frac{21}{2}$	84	169	4088	13	3960	8	0	16
3	2	4	3	4	0	4	0	–	–
3	$\frac{5}{2}$	2	1	2	0	2	0	–	–
3	3	8	5	8	0	8	0	–	–
3	$\frac{7}{2}$	4	5	8	0	8	0	–	–
3	4	13	12	21	0	21	0	–	–
3	$\frac{9}{2}$	8	8	18	0	18	0	–	–
3	5	20	24	54	0	54	0	–	–
3	$\frac{11}{2}$	13	19	62	0	62	0	–	–
3	6	31	40	149	1	147	1	0	0
3	$\frac{13}{2}$	20	32	176	1	174	0	–	–
3	7	44	69	400	4	390	2	0	0
3	$\frac{15}{2}$	31	54	478	3	468	1	0	0
3	8	61	113	1016	8	986	8	0	4
3	$\frac{17}{2}$	44	92	1304	7	1270	3	0	2
3	9	84	171	2634	19	2542	11	0	12

4. $\mathcal{N}=2$ SCQCD, $SU(N)$, $N_f=2N$

4.1 Action and components

$$V_0^{(1)}=\Big\{\begin{pmatrix}a&0\\0&0\end{pmatrix}:\operatorname{tr}a=0\Big\}=\mathfrak{su}(N),\qquad V_1=\Big\{\begin{pmatrix}0&x\\y&0\end{pmatrix}\Big\},\quad x,y\in\mathbb{C}^{N\times N}$$

$$t_A \text{ Hermitian basis of } V_0^{(1)}\oplus V_1,\quad \operatorname{tr}(t_At_B)=\tfrac12\delta_{AB},\qquad C_{ABC}=-2i\operatorname{tr}\big(t_A[t_B,t_C]\big)$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0^{(1)}}$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$$A\in V_0^{(1)};\qquad B,C\in V_0^{(1)}\ (I=3),\qquad B,C\in V_1\ (I=1,2);\qquad D,E\in V_1,\ F\in V_0^{(1)}$$

$$(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\qquad \mathsf{w}^A_{\dot a}\mapsto\tfrac12\partial_{+\dot a}\mathsf{c}^A$$

components: $\underbrace{\mathsf{c}^A,\mathsf{b}_A}_{\text{vector}}$, $\underbrace{(\gamma^3)_A,(\beta_3)^A}_{\text{adjoint chiral}}$, $\underbrace{(\gamma^{1,2})_B,(\beta_{1,2})^B,\ B\in V_1}_{2N\text{ fundamental hypermultiplets}}$; $SU(N)_2\subset U(2N)_{\text{flavour}}$ singlets

$=\kappa\to0$ limit of the $\mathbb{Z}_2$ quiver (node 2 vector multiplet decoupled)

4.2 Lie algebra realization and highest weights

$$\mathfrak{g}=\Big\{\begin{pmatrix}a&0\\0&0\end{pmatrix}:\operatorname{tr}a=0\Big\},\qquad \text{node-2 roots: Haar measure only;}\qquad G_{\mathrm{HW}}\text{ as 3.3}$$

4.3 Frontier

(all sectors)

$$SU(2),\ N_f=4:\ E+J_L\le12;\qquad SU(3),\ N_f=6:\ E+J_L\le9$$

4.4 Classes and quantum lift

Classes f: open (no stable comparison); E_2 classes per level in 4.5

$$SU(2):\ \#\{d_1\neq0\}=0\ \ (E+J_L\le12)$$
$$\operatorname{rk}d_2=0\text{ on every candidate pair},\qquad \max\operatorname{rk}d_2=0$$
$$SU(2):\ 42\text{ pairs }(Q_2\mathcal{K}\neq0:\ 41),\qquad SU(3):\ 13\text{ pairs }(Q_2\mathcal{K}\neq0:\ 5)$$
$$SU(3):\ d_1\neq0:\ (1,1,5;0,0)Y6,\ (1,1,4;1,0)Y6,\ (1,1,3;1,1)Y6,\ (1,1,6;0,0)Y6$$
$$(1,1,5;1,0)Y6,\ (1,1,4;1,1)Y6,\ (1,1,4;2,0)Y6,\ (1,1,7;0,0)Y8$$

4.5 Data

Columns

$$\text{HW blocks}=\#\{(\lambda,Y):h>0\},\qquad \textstyle\sum h=\sum_{(\lambda,Y)}h$$
$$\#\{d_1\neq0\}=\#\{(\lambda,Y):\operatorname{rk}d_1>0\},\qquad \sum E_2=\sum h-2\sum\operatorname{rk}d_1$$
$$d_2\text{ pairs}=\#\big\{(\lambda,Y):E_2^Y(\lambda)>0,\ E_2^{Y-1}(\lambda-2\mathsf{d})>0\big\}\ \ (\text{where }d_2\text{ can act})$$
$$\max\operatorname{rk}d_2=0\ \Rightarrow\ \text{no two-loop lift}$$
$$\#\{Q_2\mathcal{K}\neq0\}=\#\{\text{pairs with }Q_2\neq0\text{ on the }E_2\text{ cocycles}\}\ \ (\text{non-trivial tests})$$

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1\neq0\}$	$\sum E_2$	d_2 pairs	$\max\operatorname{rk}d_2$	$\#\{Q_2\mathcal{K}\neq0\}$
2	2	4	3	3	0	3	0	–	–
2	$\frac52$	2	1	1	0	1	0	–	–
2	3	8	3	4	0	4	0	–	–
2	$\frac72$	4	3	4	0	4	0	–	–
2	4	13	7	10	0	10	0	–	–
2	$\frac92$	8	4	6	0	6	0	–	–
2	5	20	9	21	0	21	0	–	–

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1\neq 0\}$	$\sum E_2$	d_2 pairs	$\max \mathrm{rk}\, d_2$	$\#\{Q_2\mathcal{K}\neq 0\}$
2	$\frac{11}{2}$	13	9	21	0	21	0	–	–
2	6	31	19	37	0	37	0	–	–
2	$\frac{13}{2}$	20	14	54	0	54	0	–	–
2	7	44	26	112	0	112	0	–	–
2	$\frac{15}{2}$	31	23	109	0	109	0	–	–
2	8	61	42	232	0	232	1	0	0
2	$\frac{17}{2}$	44	30	307	0	307	1	0	1
2	9	84	59	481	0	481	1	0	1
2	$\frac{19}{2}$	61	54	623	0	623	1	0	1
2	10	111	87	1263	0	1263	4	0	4
2	$\frac{21}{2}$	84	75	1527	0	1527	5	0	5
2	11	144	122	2632	0	2632	8	0	8
2	$\frac{23}{2}$	111	106	3850	0	3850	7	0	7
2	12	186	155	6440	0	6440	14	0	14
3	2	4	3	3	0	3	0	–	–
3	$\frac{5}{2}$	2	1	1	0	1	0	–	–
3	3	8	4	5	0	5	0	–	–
3	$\frac{7}{2}$	4	4	5	0	5	0	–	–
3	4	13	9	12	0	12	0	–	–
3	$\frac{9}{2}$	8	6	8	0	8	0	–	–
3	5	20	14	25	0	25	0	–	–
3	$\frac{11}{2}$	13	12	26	0	26	0	–	–
3	6	31	26	55	0	55	0	–	–
3	$\frac{13}{2}$	20	23	66	0	66	0	–	–
3	7	44	40	150	1	148	1	0	0
3	$\frac{15}{2}$	31	35	161	1	159	1	0	0
3	8	61	69	323	2	319	2	0	0
3	$\frac{17}{2}$	44	56	424	1	420	2	0	1
3	9	84	101	781	3	773	7	0	4

5. Massive-adjoint flows

5.1 Action and components

$\boxed{\mathcal{N}=4\ SU(N),\ \text{one mass}}$ $N=2,3$; $A=1,\dots,N^2-1$; $[t_B,t_C]=iC^A{}_{BC}t_A$, $\mathrm{tr}(t_At_B)=\frac{1}{2}\delta_{AB}$, $(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB}$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+\tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F+\tfrac{M}{2}\,\delta^{AB}(\gamma^3)_A(\gamma^3)_B-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$\boxed{\text{SCQCD } SU(N)+\text{mass}}$ $N=2,3$; index ranges, t_A, C_{ABC} as 4.1

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0^{(1)}}$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F+\tfrac{M}{2}\,\delta^{AB}(\gamma^3)_A(\gamma^3)_B-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$\boxed{\mathbb{Z}_2\text{ quiver }SU(2)\times SU(2)+\text{masses}}$ index ranges, t_A, C_{ABC} as 3.1; $\Lambda^{AB}=M\delta^{AB}$ ($A,B\in$ node 1), $-M\delta^{AB}$ ($A,B\in$ node 2)

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0}$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F+\tfrac{1}{2}\Lambda^{AB}(\gamma^3)_A(\gamma^3)_B-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$$Q_0'=Q_0+\Lambda^{AB}(\gamma^3)_B\frac{\partial}{\partial(\beta_3)^A},\qquad Q_n'=Q_n\ (n\geq 1),\qquad \Lambda^{AB}=M\delta^{AB}\ \text{(one node)}$$

UV	mass	IR
$\mathcal{N}=4$ $SU(2)$, $SU(3)$	M on γ^3	Leigh–Strassler; $\bar{W}_{\rm IR} = -\frac{1}{2M}\,\mathrm{Tr}[\gamma^1,\gamma^2]^2$
SCQCD $SU(2)$, $SU(3)$	M on γ^3	$\mathcal{N}=1$ SQCD, $N_f = 2N$, quartic $\bar{W}$
$\mathbb{Z}_2$ quiver $SU(2)^2$	$(M_1, M_2) = (M, -M)$	Klebanov–Witten

5.2 Gradings and charges

$$(A,B;n_+,n_-;\mathrm{gh}):\quad A=\theta_1+\theta_3,\quad B=\theta_2+\theta_3,\quad \mathrm{gh}=Y-\Theta;\qquad C(A,B;n_\pm;\mathrm{gh})=\bigoplus_{\theta_3} C^Y(\lambda)$$

$$R_1+R_3=A+\mathrm{gh},\quad R_2+R_3=B+\mathrm{gh},\quad J_L=\tfrac{1}{2}(n_++n_--\mathrm{gh}),\quad J_R=\tfrac{1}{2}(n_+-n_-)$$

$$R_{\rm IR}=\tfrac{1}{2}(R_1+R_2)+R_3,\qquad E_{\rm IR}=\tfrac{3}{2}R_{\rm IR}+2J_L,\qquad w=A+B+2(n_++n_-)=2R_{\rm IR}+4J_L=\tfrac{4}{3}\big(E_{\rm IR}+J_L\big)$$

$$d_r:\ (A,B;n_\pm;\mathrm{gh})\to(A-2r,B-2r;n_\pm+r;\mathrm{gh}+1+2r),\qquad w\ \text{fixed}$$

$$G_{\rm HW}=GL(2)_{12}\times SL(2)_{\dot a}:\quad A\ge B\ge 0,\ \ n_+\ge n_-\ge 0;\qquad \text{Lie algebra as 1.2, 3.2, 4.2}$$

5.3 Frontier

theory	complete	d_r pairs ($r=2,3$)	$\max \mathrm{rk}\, d_r$
$\mathcal{N}$=4 $SU(2)$, one mass	$w\le 16$ (partial: $w=17$)	101	0
$\mathcal{N}$=4 $SU(3)$, one mass	$w\le 13$	49	0
SCQCD $SU(2)$ + mass	$w\le 15$	27	0
SCQCD $SU(3)$ + mass	$w\le 13$	13	0
$\mathbb{Z}_2$ quiver $SU(2)^2$, masses $(M,-M)$	$w\le 13$	16	0

5.4 Classes and quantum lift

theory	class	$(A,B;n_+,n_-;\mathrm{gh})$	$(R_1{+}R_3, R_2{+}R_3)$	J_L	$R_{\rm IR}$	$E_{\rm IR}$	quantum
$SU(2)$, one mass	w	$(3,3;0,0;-2)$	$(1,1)$	1	1	$\frac{7}{2}$	graviton
	w^2	$(6,6;0,0;-4)$	$(2,2)$	2	2	7	closed completion to $\hbar^3$ (quartic $\bar{W}$)
	$\mathcal{O}_7$	$(5,2;0,0;-2)$	$(3,0)$	1	$\frac{3}{2}$	$\frac{17}{4}$	$d_1=0$; $r\ge 2$ empty $\Rightarrow$ all orders [p]
	$\mathcal{O}_{16}$	$(8,8;0,0;-5)$	$(3,3)$	$\frac{5}{2}$	3	$\frac{19}{2}$	all orders [p]

$$\mathcal{O}_7,\mathcal{O}_9,\mathcal{O}_{14},\mathcal{O}_{16}:\ \text{2512.12674}$$

$$\mathcal{O}_{16}:\ d_1\ (h_T=0),\ d_2\ (h_T=7,\ Q_2\mathcal{K}\neq 0),\ d_3\ (h_T=8),\ d_4\ (Q_4\mathcal{K}=0)=0;\ \text{no } d_5,\ \text{no incoming}$$

$$SU(2),\ w\le 17:\ d_1\neq 0 \text{ only at } (6,6;1,1;-3);\qquad \mathrm{rk}\, d_2=\mathrm{rk}\, d_3=0 \text{ on every candidate pair}$$

$$\mathbb{Z}_2 \text{ quiver}:\ d_1\neq 0 \text{ at } (2,2;k,1;1),\ k=1,2,3$$

$$\text{other theories: } d_1=0,\qquad \mathrm{rk}\, d_2=\mathrm{rk}\, d_3=0$$

Massive SQED analogue

theory	$(7,7;1,0;-5)$	$(7,7;1,0;-4)$	$(7,7;1,0;-6)$
$U(1)$, $\mathcal{I}=ig\mathsf{c}(\gamma_+\beta^+-\gamma_-\beta^-)+\frac{m}{2}\gamma_s^2+y\gamma_s\gamma_+\gamma_-$	$d_2\neq 0$ (closed form)		
$\mathcal{N}=4$ $SU(2)$, one mass	$h=0$	$h=1$, $d_2=0$ ($Q_2\mathcal{K}\neq 0$)	$h=0$
$\mathcal{N}=4$ $SU(3)$, one mass	$h=0$	running	
SCQCD $SU(2)$ + mass	$E_1=0$		

5.5 Data

Columns as 3.6 with $(\lambda,Y)\to(A,B;n_\pm;\mathrm{gh})$; d_r pairs: $E_2>0$ at source and target of d_r, $r=2,3$; all $\mathrm{rk}\, d_r=0$

theory	w	sectors	HW blocks	$\sum h$	$\#\{d_1\neq 0\}$	$\sum E_2$	d_r pairs	r tested
$\mathcal{N}$=4 $SU(2)$, one mass	4	7	4	4	0	4	–	–
	5	7	4	4	0	4	–	–
	6	13	11	11	0	11	–	–
	7	13	12	13	0	13	–	–
	8	22	25	30	0	30	–	–
	9	22	24	40	0	40	–	–
	10	34	51	81	0	81	–	–
	11	34	49	115	0	115	–	–
	12	50	93	218	0	218	4	d_2

theory	w	sectors	HW blocks	$\sum h$	$\#\{d_1 \neq 0\}$	$\sum E_2$	d_r pairs	r tested
	13	50	90	317	0	317	4	d_2
	14	70	154	569	0	569	11	d_2
	15	70	151	845	0	845	14	d_2, d_3
	16	95	240	1473	1	1471	32	d_2, d_3
	17	95	241	2194	0	2194	36	d_2, d_3
$\mathcal{N}$=4 $SU(3)$, one mass	4	7	6	6	0	6	–	–
	5	7	10	10	0	10	–	–
	6	13	19	23	0	23	–	–
	7	13	27	42	0	42	–	–
	8	22	50	96	0	96	1	d_2
	9	22	58	171	0	171	1	d_2
	10	34	98	358	0	358	5	d_2
	11	34	112	641	0	641	7	d_2
	12	50	178	1284	0	1284	15	d_2
	13	50	194	2300	0	2300	20	d_2, d_3
SCQCD $SU(2)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	16	46	0	46	–	–
	7	13	0	0	0	0	–	–
	8	22	34	194	0	194	–	–
	9	22	0	0	0	0	–	–
	10	34	62	826	0	826	2	d_2
	11	34	0	0	0	0	–	–
	12	50	109	3371	0	3371	7	d_2
	13	50	0	0	0	0	–	–
	14	70	173	13622	0	13622	18	d_2, d_3
	15	70	0	0	0	0	–	–
SCQCD $SU(3)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	18	44	0	44	–	–
	7	13	0	0	0	0	–	–
	8	22	35	203	0	203	1	d_2
	9	22	0	0	0	0	–	–
	10	34	71	934	0	934	3	d_2
	11	34	0	0	0	0	–	–
	12	50	125	4430	0	4430	9	d_2
	13	50	0	0	0	0	–	–
$\mathbb{Z}_2$ quiver $SU(2)^2$, masses $(M, -M)$	4	7	10	16	0	16	–	–
	5	7	0	0	0	0	–	–
	6	13	26	66	0	66	–	–
	7	13	0	0	0	0	–	–
	8	22	53	299	1	297	–	–
	9	22	0	0	0	0	–	–
	10	34	98	1306	1	1304	3	d_2
	11	34	0	0	0	0	–	–
	12	50	170	5442	1	5440	13	d_2, d_3
	13	50	0	0	0	0	–	–

6. $\mathcal{N} = 4$ $SU(3)$: β and Leigh–Strassler deformations

6.1 Action and components

$$A=1,\dots,8,\qquad [t_B,t_C]=iC^A{}_{BC}t_A,\quad \mathrm{tr}(t_At_B)=\tfrac12\delta_{AB},\quad (t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\quad d_{ABC}=2\,\mathrm{tr}\big(t_A\{t_B,t_C\}\big)$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+\tfrac{1+q}{4}\cdot\tfrac{i}{3!}\,\delta^{AD}\delta^{BE}\delta^{CF}\big(\varepsilon_{IJK}C_{ABC}-iS_{IJK}d_{ABC}\big)(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$$S_{IJK}=\tfrac{1-q}{1+q}|\varepsilon_{IJK}|+\tfrac{2h}{1+q}\delta_{IJK},\qquad \delta_{IJK}=1\ (I=J=K),\ 0\ \text{else}$$

$$\tfrac{1+q}{4}\cdot\tfrac{i}{3!}\,\delta^{AD}\delta^{BE}\delta^{CF}\big(\varepsilon_{IJK}C_{ABC}-iS_{IJK}d_{ABC}\big)(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F$$
$$=\mathrm{tr}\,\gamma^1\gamma^2\gamma^3-q\,\mathrm{tr}\,\gamma^1\gamma^3\gamma^2+\tfrac{h}{3}\sum_I\mathrm{tr}(\gamma^I)^3,\qquad \gamma^I=(\gamma^I)_At_A$$

$$\underbrace{q=1,\ h=0}_{\mathcal{N}=4\ (\times\frac12)},\qquad \underbrace{h=0}_{\beta},\qquad \underbrace{(q,h)\ \text{generic}}_{\text{Leigh–Strassler}};\qquad SU(2):\ d_{ABC}=0\ \Rightarrow\ \text{deformation trivial}$$

components: c^A, b_A (vector), $(\gamma^I)_A$, $(\beta_I)^A$ ($I=1,2,3$ adjoint chirals)

$$Q_0'\beta_I=Q_0\beta_I\big|_{\varepsilon C\to\varepsilon C-iSd}$$

Q_n' : fan rule, $\gamma\gamma\gamma$ vertex at slot k : $\varepsilon_{IJK}\hat A_n\,[L,R]\ +\ S_{IJK}\hat A_n\,\{L,R\}_{\mathfrak{su}(3)}$

6.2 Lie algebra realization

$$\mathfrak{su}(3)\ \text{as 1.2};\qquad \mathrm{tr}(t_At_B)=\tfrac12\delta_{AB},\qquad \{t_A,t_B\}=\tfrac13\delta_{AB}\mathbf{1}+d_{ABC}t_C$$
$$\mathrm{tr}(t_At_Bt_C)=\tfrac14\big(d_{ABC}+iC_{ABC}\big),\qquad \{L,R\}_{\mathfrak{su}(3)}=\{L,R\}-\tfrac23\,\mathrm{tr}(LR)\mathbf{1}$$

6.3 Highest-weight algorithm

$$\beta:\ G_{\mathrm{HW}}=U(1)^3_\theta\times SL(2)_{\dot a},\qquad C^Y(\theta_1,\theta_2,\theta_3;n_+,n_-)=\{\text{colour singlets}\}\cap\ker J_+\ \ \text{(no flavour raising)}$$
$$\mathrm{LS}:\ U(1)^3\to U(1)\times\mathbb{Z}_3,\qquad \mathcal{C}^Y(\Theta,c;n_+,n_-)=\bigoplus_{\theta_1+\theta_2+\theta_3=\Theta,\ \theta_2+2\theta_3\equiv c\ (3)}C^Y(\theta;n_+,n_-)$$

$$Q_n'=Q_n^{(0)}+\sum_I Q_n^{(h,I)},\qquad Q_n^{(h,I)}:\ \theta\mapsto\theta-n(1,1,1)+3e_I-(1,1,1)$$

$$d_r:\ (\Theta,c;n_\pm)_Y\to(\Theta-3r,c;n_\pm+r)_{Y+1-r}$$

conserved: $E,\ J_L,\ J_R,\ R_1+R_2+R_3=\tfrac12(3Y-\Theta),\ c$ (LS); cyclic $\gamma^1\to\gamma^2\to\gamma^3$: $c\mapsto c+\Theta$

6.4 Frontier

$$\beta,\ \mathrm{LS}:\ E+J_L\le 9\ \ \text{(all sectors, through } d_3)$$

6.5 Fortuitous classes

theory	class	$(\Theta,c;n_+,n_-)_Y$	E	$\sum R_I$	J_L	J_R	data
LS	fortuitous	$(7,0;0,0)_5$	6	4	1	0	$h^{SU(3)}=5,\ h^{SU(4,5,6)}=0$
LS	fortuitous	$(7,0;1,0)_5$	7	4	$\frac32$	$\frac12$	$h^{SU(3)}=5,\ h^{N\to\infty}=0$

$SU(N\ge Y+1)$: no trace relations at $Y\ \Rightarrow\ h_Y^{N\to\infty}$; full fortuity classification: open

6.6 Quantum lift

$d_2 \neq 0$, LS $SU(3)$ [p]; seeds, p', $(q,h) = (-\frac{3889}{4341}, \frac{976546}{421077}), (4, \frac{85}{6}), (1, \frac{4321}{97})$: $\mathrm{rk}\, d_2 = 1$; $h = 0$: $\mathrm{rk}\, d_2 = 0$

source	target	$(E, J_L, J_R) \to (E, J_L, J_R)$	$Q_2\mathcal{K}$	$\mathrm{rk}\, d_2$
$(7,0;0,0)_5$	$(1,0;2,2)_4$	$(6,1,0) \to (\frac{13}{2}, \frac{1}{2}, 0)$	0	1
$(8,0;0,0)_6$	$(2,0;2,2)_5$	$(7,1,0) \to (\frac{15}{2}, \frac{1}{2}, 0)$	0	1
$(7,0;1,0)_5$	$(1,0;3,2)_4$	$(7, \frac{3}{2}, \frac{1}{2}) \to (\frac{15}{2}, 1, \frac{1}{2})$	$\neq 0$	1

$$Q_2\mathcal{K} = 0: \quad d_2[\mathcal{O}_0] = [Q_1\mathcal{O}_1], \qquad Q_0'\mathcal{O}_1 = -Q_1'\mathcal{O}_0$$

$$(7,0;0,0)_5: \; h_{E_1} = 5 \xrightarrow{d_1,\ \mathrm{rk}\,3} h_{E_2} = 2 \xrightarrow{d_2,\ \mathrm{rk}\,1} h_{E_3} = h_{E_\infty} = 1; \qquad \Theta \not\equiv 0\ (3): \ \times 3 \text{ cyclic images}$$

β: $\mathrm{rk}\, d_2 = 0$ on all 11 candidate pairs; β, LS: no d_3 candidates $(E + J_L \leq 9)$

Gates $(Q_0')^2 = 0$, $\{Q_0', Q_1'\} = 0$, $\sum_j Q_j Q_{2-j} = 0$, $\sum_j Q_j Q_{3-j} = 0$ (nonzero terms; β and LS); controls: d-slot sign, no $\mathfrak{su}(3)$ projection, Q_2/Q_3 d-part $\times(-1)$, h-vertex $\times 2$: violations

6.7 Data

β

j	sectors	$\sum h_{E_1}$	d_1 pairs	$d_1 \neq 0$	d_2 pairs	$d_2 \neq 0$	d_3 pairs	$d_3 \neq 0$	$\sum h_{E_4}$
2	1	0	0	0	0	0	0	0	0
3	1	0	0	0	0	0	0	0	0
4	2	2	0	0	0	0	0	0	2
5	1	1	0	0	0	0	0	0	1
6	5	5	1	0	0	0	0	0	5
7	2	3	0	0	0	0	0	0	3
8	6	6	0	0	0	0	0	0	6
9	5	9	1	0	0	0	0	0	9
10	9	14	2	0	0	0	0	0	14
11	6	16	1	0	0	0	0	0	16
12	16	44	5	0	1	0	0	0	44
13	9	40	3	0	0	0	0	0	40
14	19	71	8	2	0	0	0	0	67
15	16	108	8	1	1	0	0	0	106
16	26	149	17	1	4	0	0	0	147
17	19	190	13	4	1	0	0	0	182
18	39	383	25	7	4	0	0	0	369

j	source	target	E_2	E_2	$Q_2\mathcal{K} \neq 0$	$\mathrm{rk}\, d_2$	$(E, J_L, J_R) \to (E, J_L, J_R)$
12	$(2,2,2;0,0)_5$	$(0,0,0;2,2)_4$	2	1	0	0	$(\frac{11}{2}, \frac{1}{2}, 0) \to (6,0,0)$
15	$(2,2,2;1,0)_5$	$(0,0,0;3,2)_4$	6	1	0	0	$(\frac{13}{2}, 1, \frac{1}{2}) \to (7, \frac{1}{2}, \frac{1}{2})$
16	$(3,3,2;0,0)_6$	$(1,1,0;2,2)_5$	2	1	0	0	$(7,1,0) \to (\frac{15}{2}, \frac{1}{2}, 0)$
16	$(4,2,2;0,0)_5$	$(2,0,0;2,2)_4$	1	2	4	0	$(\frac{13}{2}, \frac{3}{2}, 0) \to (7,1,0)$
16	$(4,2,2;0,0)_6$	$(2,0,0;2,2)_5$	1	2	0	0	$(7,1,0) \to (\frac{15}{2}, \frac{1}{2}, 0)$
16	$(4,2,2;0,0)_7$	$(2,0,0;2,2)_6$	1	1	0	0	$(\frac{15}{2}, \frac{1}{2}, 0) \to (8,0,0)$
17	$(3,2,2;1,0)_6$	$(1,0,0;3,2)_5$	9	1	0	0	$(\frac{15}{2}, 1, \frac{1}{2}) \to (8, \frac{1}{2}, \frac{1}{2})$
18	$(2,2,2;2,0)_5$	$(0,0,0;4,2)_4$	9	2	4330	0	$(\frac{15}{2}, \frac{3}{2}, 1) \to (8,1,1)$
18	$(3,3,3;0,0)_6$	$(1,1,1;2,2)_5$	2	2	2672	0	$(\frac{15}{2}, \frac{3}{2}, 0) \to (8,1,0)$
18	$(3,3,3;0,0)_7$	$(1,1,1;2,2)_6$	1	5	0	0	$(8,1,0) \to (\frac{17}{2}, \frac{1}{2}, 0)$
18	$(5,2,2;0,0)_7$	$(3,0,0;2,2)_6$	3	2	0	0	$(8,1,0) \to (\frac{17}{2}, \frac{1}{2}, 0)$

LS

j	classes	$\sum h_{E_1}$	d_1 pairs	$d_1 \neq 0$	d_2 pairs	$d_2 \neq 0$	d_3 pairs	$d_3 \neq 0$	$\sum h_{E_4}$
2	1	0	0	0	0	0	0	0	0
3	2	0	0	0	0	0	0	0	0
4	1	2	0	0	0	0	0	0	2
5	1	1	0	0	0	0	0	0	1
6	6	3	0	0	0	0	0	0	3
7	1	3	0	0	0	0	0	0	3
8	3	6	0	0	0	0	0	0	6
9	6	11	1	0	0	0	0	0	11
10	3	9	0	0	0	0	0	0	9
11	3	14	1	0	0	0	0	0	14
12	12	56	3	1	0	0	0	0	54
13	3	26	2	0	0	0	0	0	26
14	6	54	3	1	1	1	0	0	46
15	12	136	6	3	1	0	0	0	118
16	6	88	6	1	1	1	0	0	84
17	6	143	5	2	2	1	0	0	121
18	20	460	14	9	3	0	0	0	382

j	source	target	E_2	E_2	$Q_2\mathcal{K}\neq 0$	$\mathrm{rk}\, d_2$	$(E,J_L,J_R)\to(E,J_L,J_R)$
14	$(7,0;0,0)_5$	$(1,0;2,2)_4$	2	2	0	1	$(6,1,0)\to(\frac{13}{2},\frac{1}{2},0)$
15	$(6,0;1,0)_5$	$(0,0;3,2)_4$	2	1	0	0	$(\frac{13}{2},1,\frac{1}{2})\to(7,\frac{1}{2},\frac{1}{2})$
16	$(8,0;0,0)_6$	$(2,0;2,2)_5$	3	3	0	1	$(7,1,0)\to(\frac{15}{2},\frac{1}{2},0)$
17	$(7,0;1,0)_5$	$(1,0;3,2)_4$	2	4	2112	1	$(7,\frac{3}{2},\frac{1}{2})\to(\frac{15}{2},1,\frac{1}{2})$
17	$(7,0;1,0)_6$	$(1,0;3,2)_5$	3	1	0	0	$(\frac{15}{2},1,\frac{1}{2})\to(8,\frac{1}{2},\frac{1}{2})$
18	$(6,0;2,0)_5$	$(0,0;4,2)_4$	4	2	5680	0	$(\frac{15}{2},\frac{3}{2},1)\to(8,1,1)$
18	$(9,0;0,0)_7$	$(3,0;2,2)_6$	8	4	0	0	$(8,1,0)\to(\frac{17}{2},\frac{1}{2},0)$
18	$(9,1;0,0)_7$	$(3,1;2,2)_6$	6	3	0	0	$(8,1,0)\to(\frac{17}{2},\frac{1}{2},0)$

N-dependence of h_{E_1} (LS)

class	Y	$SU(3)$	$SU(4)$	$SU(5)$	$SU(6)$	$SU(7)$
$(7,0;0,0)$	5	5	0	0	0	0
$(7,0;0,0)$	6	0	16	0	0	0
$(7,0;0,0)$	7	0			11	23
$(1,0;2,2)$	4	2	2		2	
$(8,0;0,0)$	6	4	11	2		2
$(2,0;2,2)$	5				8	
$(7,0;1,0)$	5	5	0	0	0	
$(7,0;1,0)$	6	10	41	9	9	
$(1,0;3,2)$	4				4	

7. Running

item	computation
$SU(3)$ $(5,4,4;1,1)Y10$	d_2 chain: $Q_1\mathcal{O}_0$, $Q_2\mathcal{O}_0$, solve $\mathcal{O}_1$, $Q_1\mathcal{O}_1$, solve R_2 (generator split)
$SO(11)$ $(7,3,3;0,0)Y12$	d_1, d_2, d_3
$SO(7)$ vs $Sp(3)$	E_2, d_2, d_3, all sectors $E+J_L\leq 12$; E_2 per (E,R_I,J_L,J_R)
$SO(7)$ β	d_1 from $(7,2,2;1,1)Y10$ (recheck), chains
SO/Sp scans	$Sp(3..5)$, $SO(11)$: $E+J_L\in[\frac{25}{2},14]$
$SU(4)$, $SU(5)$	last shapes at $E+J_L=14$, $\frac{29}{2}$
quiver, SCQCD	$SU(2)^2$ to $E+J_L=14$, $SU(3)^2$ to 10, SCQCD $SU(2)$ 12; d_3 scans
massive flows	w scans (all five theories); $SU(3)$ one mass $(7,7;1,0;-4)$ d_2
$SU(3)$ β, LS	$E+J_L\in[\frac{19}{2},11]$, d_2, d_3; operator form of the lifted pairs